

# **Business Performance Analysis and Service Optimization: A Primary Data Study of a Professional Laundry Business.**

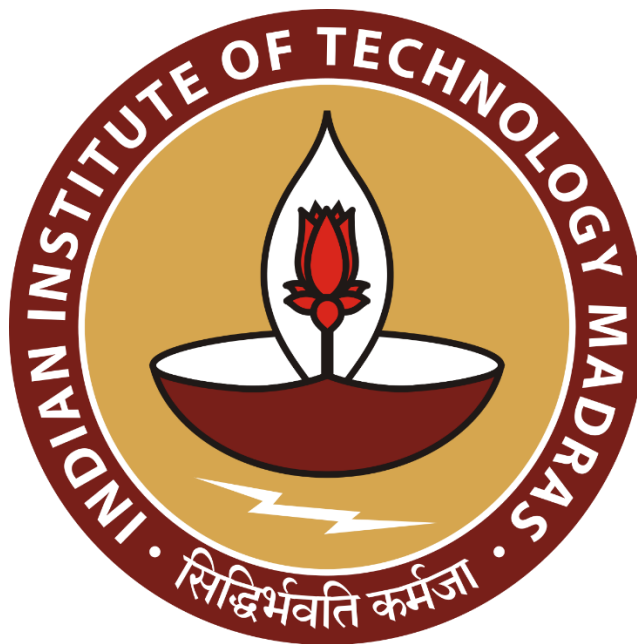
**FINAL SUBMISSION FOR THE BDM CAPSTONE PROJECT**

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**(BS) DEGREE IN DATA SCIENCE AND APPLICATIONS**

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# 1. Executive Summary

RDS Associates, operating under the trade name 'Rinz n Dry' is a privately owned B2C dry-cleaning and laundry service business operating since 2022, at MM Towers, Fatima Nagar, East Fort, Thrissur, Kerala. This Business provides multiple garment services like wash and iron, wash and fold, dry cleaning for various types of garments, shoe cleaning, bag cleaning, curtain cleaning, and blanket washing with pickup and delivery.

This study analyses 2,607 primary customer order records collected directly from the business from 10 March 2026 to 10 July 2026 – spanning four months across 106 working days, addresses three problem statements: (i) revenue and service demand imbalance, (ii) operational efficiency and turnaround time (TAT), and (iii) customer retention and behavioural patterns. The dataset was organized, cleaned and all the analysis were done using Excel and Python (Pandas). This final submission extends the mid-term analysis with additional service, time and customer level analyses - an ABC revenue classification, a revenue-density and service-level comparison, an hour by weekday demand grid, a monthly complaint trend, a customer-tier Pareto, and an acquisition-cohort retention study to analyse each problem statements.

The analysis shows Wash & Iron is the dominant service, with 47.8% of all orders (1246 of 2607) and 62.9% of total revenue (Rs. 8,24,685 of Rs. 13,11,105), shows it is an important factor for capacity and resource planning. TAT is primarily determined by the service policy, not order size. Normal-service takes 4 days, (7 days for blanket washing and shoe cleaning), while Express orders are completed in exactly 1 day. The correlation between order size and TAT is weak and even slightly negative ( $r = -0.165$ ,  $p < 0.001$ ).

Regular customers (46.4% of customer base) contribute 78.4% of all orders and 79.03% of revenue (approx. Rs. 10,36,195 of Rs. 13,11,105) average spending per order does not significantly differ between repeat and single-order customers (Rs. 507 vs Rs. 488,  $p = 0.166$ ). The higher revenue comes from purchase frequency, not larger transactions.

Complaints were recorded for 14.4% of orders (376 of 2,607), of these, 90.7% were resolved through rectification, compensation or refund. Weekly and monthly revenue is almost stable across the study period.

Based on these findings, this report recommends prioritising Wash & Iron capacity, improving TAT communication, reviewing the causes of complaints and convert first-time customers into regular customers to grow its regular-customer base rather than increasing basket size.

## 2. Detailed Explanation of Analysis Process and Methodology

### 2.1 Data Source and Collection

The project uses Primary data for the study. The data set was collected directly from the organization through an interview with the business owner, Ms. Shanitha. A written permission was attained from the business organization for the academic use. The data is spanned mainly in two categories:

First one is the **Operational data** which was collected through the direct interview, contains information on average daily revenue, staff strength, working hours, service-wise turnaround-time policies, and the full price list of different services.

Second one is the **Customer Orders data**. The significant data for the study came from this category. This data has been exported directly from business's digitalized data storage. This contains 2,607 customer order records and covers a period from 10 March 2026 to 10 July 2026, (106 Monday-Saturday working days), containing 16 columns and includes information on order timing, customer type, service type, no of garments, bill amount, payment mode, delivery type, delivery date, TAT and complaint status. All the variables are used for analysis.

### 2.2 Analytical Sequence

The analysis followed a staged sequence. First the raw data set was cleaned and done basic analysis (§2.3). Second, descriptive summaries and service-wise tabulations were done to establish the structure of the business; This stage were completed in midterm submission. Third, hypothesis testing, correlation analysis and preliminary weekly time-series analysis were done to identify patterns - also completed in midterm. Finally, each problem statements were re-examined from a different analytical angle- revenue concentration and pricing structure for PS-1, service-level and time-grid analysis for PS-2, and cohort- and tier-based segmentation for PS-3

## 2.3 Data Cleaning, Preprocessing and Validation

Before any analysis was performed, the raw data was cleaned and validated using Excel and Python (Pandas).

| Validation step                  | Result                                                                                                                                                     | Why it matters                                                                                                 | Linked PS        |
|----------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------|------------------|
| Duplicate detection              | No duplicate found in SL.No. or combination of Customer ID, Order Date, Bill Amount                                                                        | To avoid over-counting orders, skewed statistical results and false conclusion                                 | PS-1, PS-2, PS-3 |
| Missing value check              | No missing values were found in the major operational fields. Blank Complaint Status indicated “no complaints registered” and was confirmed with the owner | Incomplete data gives can ruin results and distinguish actual missing values from not-applicable value entries | PS-1, PS-2, PS-3 |
| Range and Date consistency check | Order date and delivery dates were within study window, so Sunday orders and delivery dates does not precede order date                                    | Protects TAT and time-window calculations                                                                      | PS-2             |
| Outlier Screening                | Some order fells outside IQR range. Confirmed with owner that, these are bulk transactions, not data entry errors                                          | Identify unusual values and determine whether they are genuine values or data entry errors.                    | PS-1, PS-2       |
| Date-time conversion             | Order Date, Order Time and Delivery Date were converted into date-time objects                                                                             | Support TAT calculation, hourly aggregation, weekday grouping, and weekly time series.                         | PS-2             |
| Aggregate cross-verification     | Monthly order counts and totals revenue cross- checked with the shop’s manual register and with the owner.                                                 | Confirm digital records represents original operations.                                                        | PS-1, PS-2       |

*Table 1: Validation steps and their analytical purpose*

## 2.4 Variable Modification and Derived Measures

Beyond the variables derived in midterm, (Order hour, Weekday, first to second visit gap, Consecutive visit gap, Complaint indicator, Complaint resolution group) following additional measures were derived for this analysis.

| Derived measure                  | Construction                                                                                                              | Analytical purpose                                                                                    |
|----------------------------------|---------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------|
| Order hour                       | Hour extracted from Order Time.                                                                                           | Identify high and low ordering hours in a day.                                                        |
| Weekday                          | Day of week extracted from Order Date.                                                                                    | Identify day-wise order pattern.                                                                      |
| First to second visit gap        | Number of days between a customer's first and second order.                                                               | Estimate the customer retention.                                                                      |
| Consecutive visit gap            | Number of days between each pair of consecutive orders for repeat customers.                                              | Analyse repeat-visit frequency.                                                                       |
| Complaint indicator              | Complaint Status treated as complaint when a complaint outcome is recorded.                                               | Analyse complaint incidence.                                                                          |
| Complaint resolution group       | Complaints grouped into resolved (rectification, compensation, or refund) and pending/escalated categories.               | Measure complaint resolution performance.                                                             |
| Revenue per garment              | Service-wise revenue divided by service-wise garment count.                                                               | Compare pricing efficiency across services independent of order volume (PS-1).                        |
| ABC revenue class                | Services ranked by revenue and classified by cumulative revenue share ( $A \leq 70\%$ , $B \leq 90\%$ , $C \leq 100\%$ ). | Identify high-medium-and low revenue services (PS-1).                                                 |
| Order-hour $\times$ weekday cell | Cross-tabulation of Order Hour and Weekday (66 cells).                                                                    | Identify peak staffing windows instead of hour and day considered separately (PS-2)                   |
| Monthly complaint rate           | Complaints divided by orders, grouped by calendar month.                                                                  | Check whether complaint incidence is trending or stable over the four months (PS-2).                  |
| Customer order-frequency tier    | Customers grouped into 1, 2-3, 4-6, and 7+ total orders.                                                                  | Show how revenue concentrates across the full spectrum of loyalty, not only regular vs single (PS-3). |
| Acquisition cohort               | Calendar month of a customer's first order.                                                                               | Track whether the repeat rate of newly acquired customers is changing over time (PS-3).               |

*Table 2: Variables derived and their analytical purpose*

## 2.5 Analytical Methods

### 2.5.1 Baseline methods

Descriptive statistics and Service-wise tabulations, hypothesis testing (independent two sample t-test, Chi-square test, One-way ANOVA), Pearson Correlation Analysis with significance (p-values), Preliminary weekly time series trend were done. Their outputs are only as the baseline referred to in §3.1

### 2.5.2 ABC/Pareto Classification

Services were ranked by revenue from highest to lowest and their cumulative revenue percentage was calculated. Then the services were divided into classes based on the standard ABC inventory-

classification convention, where Class A - cumulative percentage up to 70%, Class B was 70–90%, and Class C was 90–100%. This method was applied to answer PS-1 by identifying services that contribute more revenue and comparing it with service demand.

### **2.5.3 Two-Dimensional Cross-Tabulation (Demand Grid)**

The number of orders was cross-tabulated by Order Hour (9 columns for service hour) and Weekday (6 rows) and aggregated as a 6×11 demand grid. This addresses PS-2 as it integrated the single peak hour and weekday reported at mid-term into a single peak by work schedule.

### **2.5.4 Cohort and Tier Segmentation**

Customers were segmented into two ways for PS-3. First, by their total order count in four segments, to see how revenue concentrate gradually rather than as a single regular/single split. Second, customers were grouped by the month of first-order (acquisition cohorts) to see whether customers acquired later in the study window return at the same rate as those acquired earlier.

## **2.6 Tools Used**

**Microsoft Excel:** Data entry, cleaning, and organisation; descriptive formulas, Pivot Tables for service-wise and customer-wise summarisation; bar charts, pie charts, and trend lines for visualisation, ABC/Pareto, cross-tabulation

**Python** (Pandas, Numpy, Matplotlib, Scipy): data cleaning, duplicate and outlier check, variable derivation, chart generation, correlation analysis with significance testing, hypothesis tests (t-test, chi-square, ANOVA), weekly time-series tabulation, customer tier and cohort segmentation

## **3. Results and Findings**

### **3.1 Results from Mid-Term**

The mid-term submission identified the descriptive measures of the business, based on the 2,607 customer order data set. Wash & Iron dominates the business by 47.8% of all orders (62.9% of revenue), indicates its importance for capacity planning. Turnaround time was determined by service policy (4 days Normal, 1 day Express, 7 days for blanket washing/shoe cleaning) and only weak dependence on the order size ( $r=-0.165$ ,  $p < 0.001$ ); UPI is the major payment method (52.9%). Customer retention was a major revenue driver while 46.4% of customers are regular, generating 78.4% of total orders and 79% of total revenue. Overall, 14.4% of orders had complaint status with 90.7% was resolved.

### 3.2 Revenue and Service Demand Imbalance (PS-1)

From the mid-term analysis it was found that which service generates the highest revenue. The gap that remained was the structured way for deciding which services requires more investment, and whether average bill amount hides variation in per-garment cost efficiency. Following analyses addresses these questions.

#### 3.2.1 ABC/Pareto Classification of Services

A single ranked table shows which service generates more revenue, but not how many services collectively service priority treatment. The ABC method is converting this into a decision rule, like which of the services justify the dedicated capacity (A), which of the services are worth to monitor (B), and which of them not included and remain as is (C). This decision making addressing the PS-1

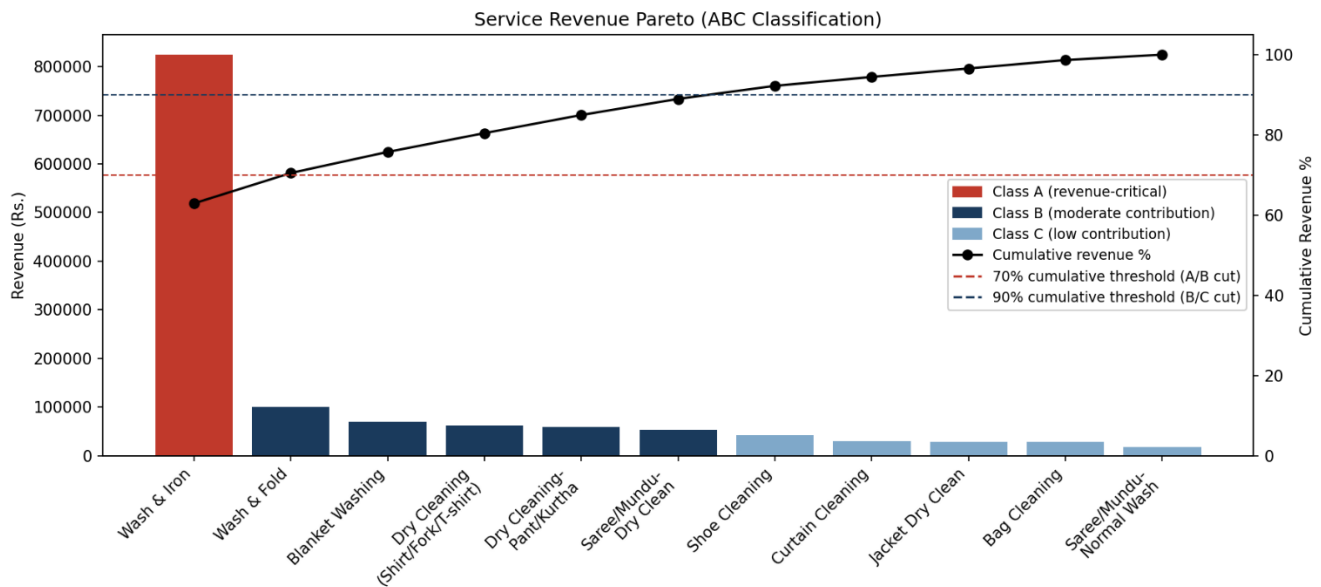


Fig 1: Service revenue Pareto with ABC classification

| Class | Services (No. of services)                                                                     | Orders | Order Share | Revenue      | Revenue Share |
|-------|------------------------------------------------------------------------------------------------|--------|-------------|--------------|---------------|
| A     | Wash & Iron (1)                                                                                | 1,246  | 47.8%       | Rs. 8,24,685 | 62.9%         |
| B     | Wash & Fold, Blanket Washing, DC-Shirt/Fork/T-shirt, DC-Pant/Kurtha, Saree/Mundu Dry Clean (5) | 959    | 36.8%       | Rs. 3,41,700 | 26.1%         |
| C     | Shoe Cleaning, Curtain Cleaning, Jacket Dry Clean, Bag Cleaning, Saree/Mundu Normal Wash (5)   | 402    | 15.4%       | Rs. 1,44,720 | 11.0%         |

Table 3: ABC classification of services by revenue



This visual representation is used because the bar and cumulative line Pareto chart makes the 70% / 90% cut point. This is the standard visual representation for the ABC decision because it shows both the individual contribution to the business (bars) and the total business (line).

Wash and Iron is top services of the business including only it as Class-A services. This service is contributing 62.9% of the total revenue from the 47.8% of total order volume. Hence considering this as the major service of the business, the staffing, machine allocation, capacity decision should be done based on this. Class-B has 5 services which is generating 26.1% of the revenue, is a monitoring group. This looks individually small but collectively over a quarter of revenue. In this Blanket Washing has only low volume of 3.7% of the orders but carries the highest bill in the shop. The Class-C also include 5 services and generate 11.0% revenue from 15.5% of total orders. Looking proportionally these services cost more staffing and chemical time per rupee generated and are best for bundling or price revision rather than dedicating to capacity.

### 3.2.2 A Revenue Density: Revenue per Garment by Service

The big multi garment order and a small single garment order can generate the same average bill for very different amounts of processing effort. Garment count effect is removed by the revenue per garment and shows which services are effectively priced as per the unit of work. This distinction is not done in the Mid-term Submission which is relevant to the PS-1.

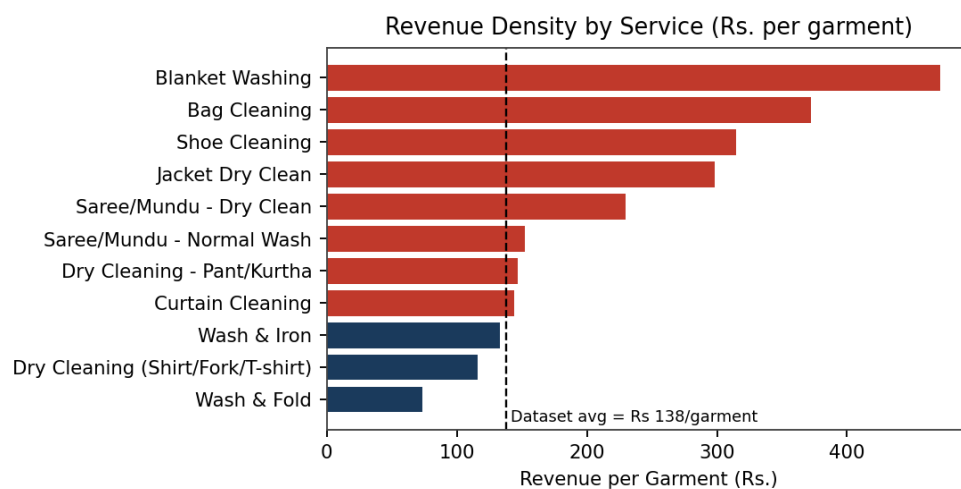


Fig 2: Revenue per garment by service, against the dataset average (Rs. 138.1/garment)

This is a horizontal bar graph for a single vertical reference line (the average of dataset). This allows to read every service as above or below par in one look, which is more suitable to read the eleven numbers in a table for the one dimensional comparison.

Blanket Washing (Rs. 471 per garment) and Bag Cleaning (Rs. 373 per garment) are more effectively priced services. Wash and Fold has the lowest pricing comparing to the other services which is Rs. 73 per Garment. This is very lower than the average price of Rs. 138 per garment. At the same time this is the second and largest service by the order volume (12.8%). Coming to the pricing, Wash and Iron is near to the average bill (Rs. 133 per Garment). This is not only because of the smaller order size but also the low per garment price, which is a pricing lever rather than a demand pattern analysis.

### 3.2.3 Service Preference by Customer Type - New vs Regular

This section shows whether particular service act as an acquisition chance like disproportionately showing more new customers or a loyalty channel like disproportionately showing more regular customers. This will help to understand whether the market effort should be same for all service menu or targeted at specific services. This is directly addressing the PS-1 demand imbalance from the customer-mix angle rather than the revenue angle.

| Service                   | New orders | Regular orders | New customer % |
|---------------------------|------------|----------------|----------------|
| Blanket Washing           | 47         | 50             | 48.5%          |
| Bag Cleaning              | 32         | 42             | 43.2%          |
| Curtain Cleaning          | 44         | 57             | 43.6%          |
| Wash and Iron             | 502        | 744            | 40.3%          |
| Saree/Mundu - Normal Wash | 31         | 46             | 40.3%          |
| Wash and Fold             | 131        | 203            | 39.2%          |
| Saree/Mundu - Dry Clean   | 49         | 71             | 40.8%          |
| Shoe Cleaning             | 34         | 55             | 38.2%          |
| DC - Shirt/Fork/T-shirt   | 80         | 126            | 38.8%          |
| DC - Pant/Kurtha          | 78         | 124            | 38.6%          |
| Jacket Dry Clean          | 22         | 39             | 36.1%          |

*Table 4: New vs Regular customer share by service*

In overall dataset 40.3 % of the share are from new customers. From the table it is understood that each services fall within or about  $\pm 8$  percentage points. Blanket Washing is acting as the most first timers which is 48.5 % and the Jacket Dry Cleaning as the lowest which is 36.1%, which the modest spread. This is a use full finding in a way that, no service of the business is representing as

a strong walk in acquisition magnet or a strong loyalty only services, so a single business wide marketing and loyalty approach is the more defensible use of a small marketing budget.

### 3.3 Operational Efficiency and Turnaround Time (PS-2)

Midterm submission incorporated the TAT by service and the weak TAT by the order volume. This analysis is solving the gaps in the previous analysis like not examine the Normal vs Express as a business decision, did not integrate the hour and weekday views into staffing grid, and did not check whether complaint incidence is charging over time.

#### 3.3.1 Normal vs Express Service Level – TAT and Pricing Comparison.

From the previous analyses it is understood that Normal orders take 4 days and Express order will only take 1 day to delivery as per the description policy. It did not test whether Express is also different, statistically confirmed pricing tier. This is matters for the PS-2. If the express already commands a premium with no another TAT variability, it should be understood as underused revenue lever rather than only a convenience option.

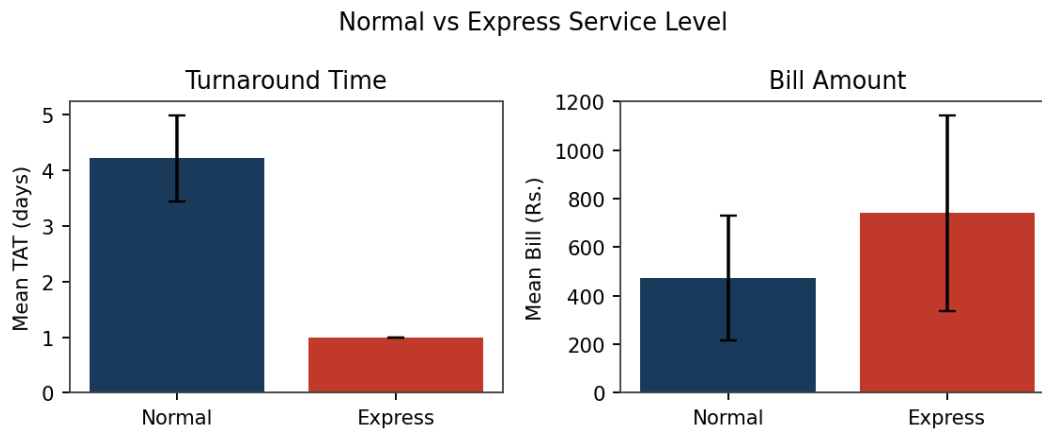


Fig 3: Mean TAT and mean bill amount, Normal vs Express (error bars = 1 SD)

| Service Level | n             | Mean TAT (days) | TAT SD | Mean Bill (Rs.) | Share of Revenue |
|---------------|---------------|-----------------|--------|-----------------|------------------|
| Normal        | 2,309 (88.6%) | 4.22            | 0.78   | 472.4           | 83.2%            |
| Express       | 298 (11.4%)   | 1.00            | 0.00   | 739.5           | 16.8%            |

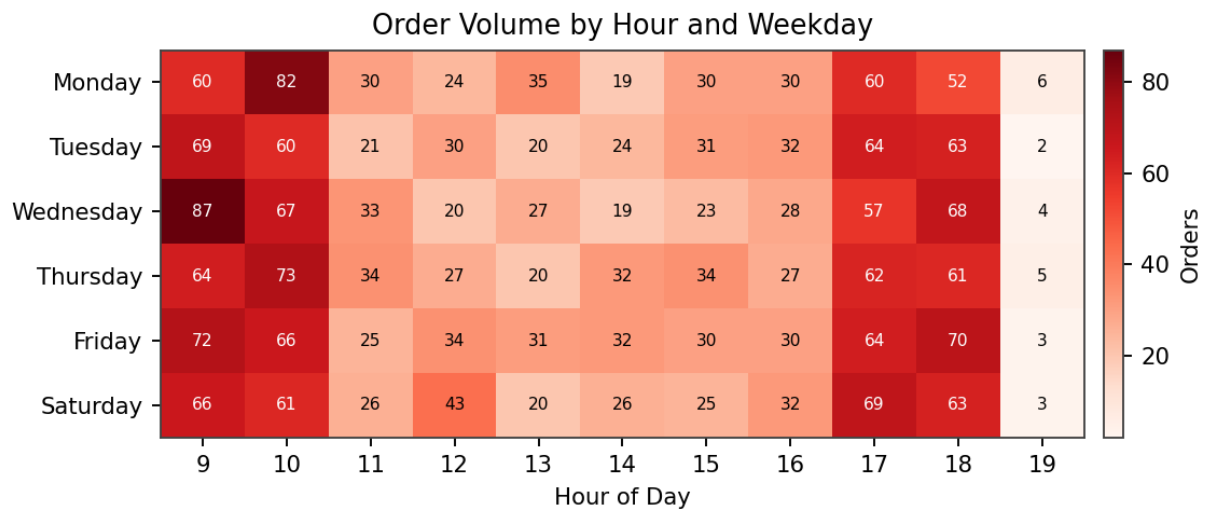
Table 5: Normal vs Express – TAT and billing comparison

The Express order contains a 56.5% higher average bill than the normal orders confirmed by an independent two sample t test (Rs. 739.5 vs Rs. 472.4,  $t = -11.12$ ,  $p < 0.001$ ). Combined with Express TAT's zero variance, this confirms Express is a premium service with 50% extra charge

that to completed by next day. This is generating 16.8% of profit from only 11.4% of total order volume. The zero variance is showing for the express TAT which means every order is delivering in exact one day. This is confirming that it is deterministic, and timely delivering premium service. Hence express is not just a speed option, it is proven and little bit under marketed premium revenue generator.

### 3.3.2 Hour × Weekday Demand Heat-Map

A Staff roster needs to understand which specific day which specific hour is busiest, not the two facts independently or separately. The single busiest day can be fit in any of the working days. The cross tabulation will specifically help to identify this and answer to the combined question of PS2.



*Fig 4: Order volume by hour of day and day of week (10 Mar – 10 Jul 2026)*

This is a heat map which visually representing 66 cell values. The darkest cells indicating the busiest cells in seconds. And the lighter shade cells indicates the least busy cells in seconds.

From this figure it can be understood that, the single busiest day in a week is Wednesday between 9-10AM. This hour is received average order of 87. Monday is the second largest busy day in the week, but between 10-11 AM and contains 82 average orders. This is showing the sharper and specific peak rather than simply saying Friday is the busiest day and roughly even by hour as per the initial findings. The two peak hour groups (9-10 Am and 5-6 Pm) are consistently dark among all the six weeks. However Wednesday and Monday mornings are showing relatively busier than Tuesday and Saturday Mornings. Hence this is showing the staff allocation should not follow similar way in all six working days. An extra allocation of staff is necessary for Wednesday and

Monday mornings. This can address the cell level demand that didn't reveal on the initial phase of study.

### 3.3.3 Monthly Complaint Rate Trend

This phase of Analysis is checking whether the complaint rate is increasing, decreasing or stable in the four month study window. This is the further analysis from the initial study which found the complaint rate is 14.4%. If the complaint rate is increasing, it indicate the problem on business's operational efficiency over time. If the complaint rate reveals stable or falling indicate the steady-state quality level of the business. This is directly extending PS-2 operational efficiency question.

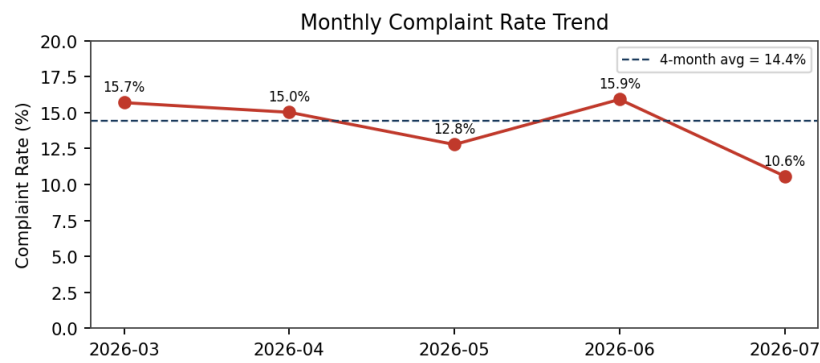


Fig 5: Complaint rate by calendar month, against the 4-month average (14.4%)

From this figure it is showing that there is not clear upward or downward trend for the complaint rate across the four month window. In the month of March it is 15.7%, April - 15.0%, May - 12.8%, June - 15.9% (The month July is excluded from the analysis because there is only 9 partial working days and 92 customers- too low numbers to make a good analysis and a chance to get a false result). From this May is the best performing month and June is the not well performed month among the four, and a 3.1 percentage point spread is there. The absence of a trend is showing it self-informative. From this it is indicating the stable operational efficiency of the business rather than decline or worsening. However, the slight increase in June above the complaint rate of March after the improvement in May signalling the dip in May was not a permanent fix and is worth a short root cause check. For example, the changes in the staffing and machine downtime etc. before June's or a similar month's trend might be repeating again. (

On the delivery surcharge revenue, an overall supporting check was also made. This is the extra distance orders are chargeable when it is beyond the free 3.5 km radius. There are total 274

chargeable orders. This shows negligible or statistically insignificant correlation with the amount of bill ( $r = -0.054$ ,  $p = 0.373$ ) and generate only Rs 6,644 or 0.5% of the total revenue. This analysis is confirming that the current distance based chargeable delivery policy is immaterial to the revenue generation and does not require to be re-engineered for that purpose.

### 3.4 Customer Retention and Behavioural Patterns (PS-3)

Midterm submission established the overall regular customer share is 46.4% and the median of first to second visit gap. It took retention as a two value variable (single vs regular) and did not analyse full range of revenue concentration across customer loyalty levels, whether new customers are retaining as effectively as the older ones or whether a particular customer's complaint affects their next visit. The following analyses will address the gaps.

#### 3.4.1 Customer Order-Frequency Tiers and Revenue Concentration.

The midterm analysis just splits all the customers into two – single (53.6%) and regular (46.4%) whether they made 2 order or 14. Instead of splitting into two groups, splitting into more finer tiers based on number of order like: single, 2-3, 4-5, 7+, shows how much customers are loyal to the business and can able to target loyalty programmes, which will address PS-3 at a higher level than previous binary split analysis.

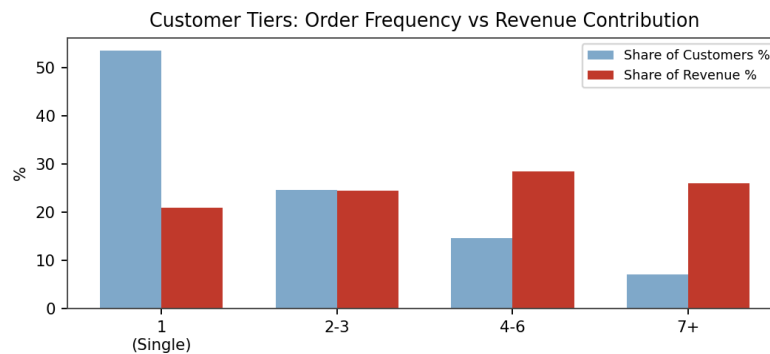


Fig 6: Share of customers vs share of revenue, by order-frequency tier

| Tier             | Customers | Customer Share | Orders | Revenue Share |
|------------------|-----------|----------------|--------|---------------|
| 1 (single order) | 563       | 53.6%          | 563    | 21.0%         |
| 2 - 3 orders     | 259       | 24.7%          | 622    | 24.5%         |
| 4 - 6 orders     | 154       | 14.7%          | 744    | 28.5%         |
| 7+ orders        | 74        | 7.0%           | 678    | 26.0%         |

Table 6: Customer order-frequency tiers and revenue share

Customers with greater than 4 orders (4-6 tier and 7+ tier) are generating 54.5% (28.5 % + 26.0%) of total revenue whose are just 21.7% (14.7% + 7%) of total customers. This shows more than half of revenue is came from roughly one in five of customers. Top 20% of customers in terms of revenue generated, level of concentrating that single vs repeat split failed to show. This means the best form of retaining customers is not from single to 2-3 tier (still an important §3.4.3) but growing 2-3 tier (259 customers 24.5% of revenue, second largest customer category) into 4-6 tier.

### 3.4.2 Acquisition-Month Cohort Repeat-Rate Analysis

However, the midterm analysis failed to show whether recently acquired customers are retained compared to previously acquired customers. The typical approach to identify such drift in customer retention is to use a cohort analysis method where the customers are grouped by the month of first order and analyse how many of them returned.

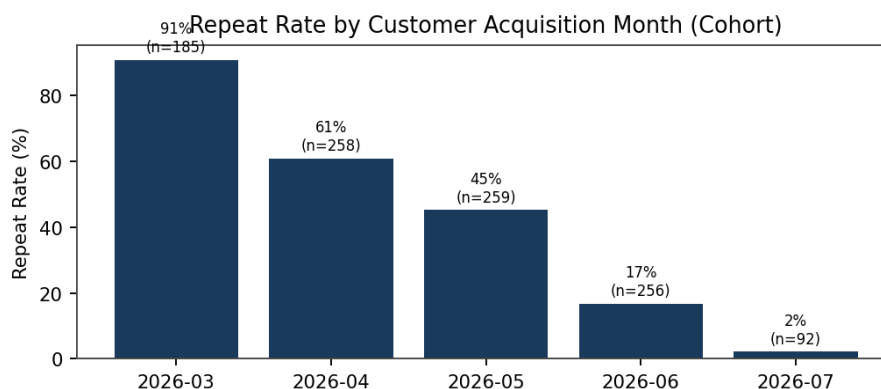


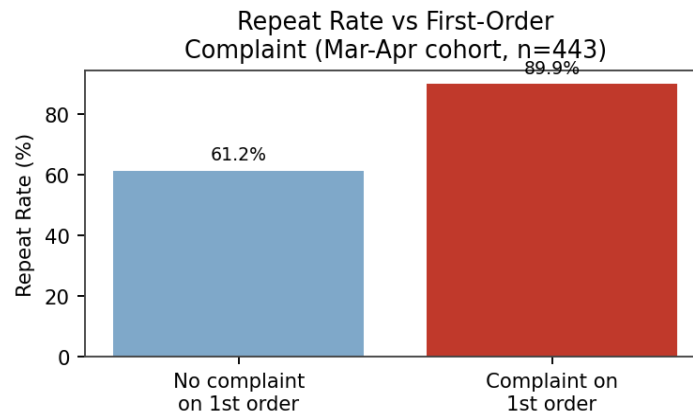
Fig 7: Repeat rate by customer acquisition month

The repeat rate decline sharply by month to month 90.8% in March cohort → 60.9% in April → 45.2% in May → 16.8% in June cohort (July cohort is excluded because of low data points). The recently acquired customers in the 4-month period simply had less time left to place another order before the data was extracted (5 weeks in June cohort vs 17 weeks in march cohort). So this comparison can made reliably only between march and April cohort, where both is having a multi-month runway. Even with these there is still a ~30% decline in repeat rate (90.8% to 60.9%), which cannot be explained by this observation period and shows importance to target first to second visit programme to recently acquired customers.

### 3.4.3 Complaint Resolution and Repeat Behaviour

The earlier analysis treated the complaint rate (PS-2) and customer retention (PS-3) are 2 distinct findings. This analysis was performed to test the connection between them; - how the upcoming

orders affect both new customers with complaint in first order and customers without complaint in their first order. This will address the connection between PS-2 and PS-3.



*Fig 8: Repeat rate by first-order complaint status  
(March–April acquisition cohort, n = 443, to avoid the right-censoring in §3.4.2)*

Discussion: restricting ourself into the March-April cohort (which has enough runway for a fair repeat rate comparison), customers with a complaint on their first order returned at a significantly higher rate (89.9%) than customers without complaint (61.2%). This was confirmed by a chi-square test ( $\chi^2 = 44.21$ ,  $p < 0.001$ ). Give that 90.7% percentage of complaints was resolved by rectification, compensation, or refund, shows that the resolvent system is working well and builds a trust and service commitment, converting a service mistake into something positive. It is truly a counter-intuitive finding: RDS Associates should not treat its complaint resolution system as some cost centre and reduce it, but as a retention mechanism to be protected.

## 4. Interpretation and Recommendations

Recommendations in §4.1 - §4.3 are data driven recommendations comes from the analysis in §3 and §4.4 are strategic recommendations comes from general customer experience and overall business goals and market trends rather than a specific pattern found in the data set.

### 4.1 Problem Statement 1 – Revenue and Service Demand Imbalance

#### Interpretation :

- Wash and Iron is a one-service Class A, contributing 47.8% of order volume and 62.9% of business's revenue. Since this is the core service of the business and should be treated as the root around resources like staff and capacity are arranged, and the ABC analysis (§3.2.1) shows no second service comes close; Class B is 5 collective services, not one.



- Revenue per garment analysis (§3.2.2) shows that Wash and Fold is priced about half of the average per garment in the dataset (Rs. 73 vs Rs. 138), even though it is the second-largest service by volume, which shows that it is a pricing gap not a demand gap.
- New customer share is almost same across all eleven services (36% - 49%), meaning no single service stands out as a main attraction for new customers (§3.2.3), which implies that targeted marketing efforts focussed on specific services are not necessary.

#### **Recommendations :**

- Allocation of Staff and Machines on daily basis especially for Wash and Iron services as per their higher demand and high contribution, treating it as a single Class-A. Schedule Class-B and Class-C services around the availability of Class A capacity, not treated as equal priorities. *[Data driven - §3.2.1 ABC classification, PS-1]*
- Review the pricing or garment-count range for Wash & Fold service specifically - either by implementing small increase in per-garment price or by a minimum garment-count requirement; since its current rate (Rs. 73/garment) is significantly lower than the dataset average (Rs. 138). *[Data driven - §3.2.2 revenue-density analysis, PS-1]*
- Maintain customer-acquisition marketing such as offers, signage, referral cards, etc; across all services rather than focusing them just one or two services, since no service shows a clear new customer advantage. *[Data driven - §3.2.3 new-vs-regular cross-tabulation, PS-1]*

## **4.2 Problem Statement 2 – Operational Efficiency and Turnaround Time**

#### **Interpretation :**

- Express service takes exactly 1 day with no variation in turnaround time. Its average bill is 56.5% higher than Normal orders, but it makes up for only 11.4% of all orders. This shows that (§3.3.1) Express is a reliable premium service, but is currently used by few customers only.
- The busiest time is on Wednesdays between 9-10 AM, with 87 orders, followed by Mondays between 10-11 AM, with 82 orders. Although 9-10 AM and 5-6 PM are the primary peak times, the level of demand is not consistent across all weekdays (§3.3.2).
- The average complaint rate in the business is 14.4%, and is stable across the study period. However, complaints rise from 12.8% in May to 15.9% in June. The supporting analysis

(§3.3.3) also found that, delivery-distance surcharges contributed only 0.5% of total revenue, making them a minor revenue factor.

**Recommendations :**

- Promote the Express service more effectively through clear pricing labels, signage at counters, and straightforward upselling during order process. This service offers a reliable one-day turnaround and generates higher average revenue. *[Data driven - §3.3.1 Normal vs Express comparison, PS-2]*
- Rather than spreading staffs and resources evenly across the whole day, there should be allocation of staffs and resources around 9-10 AM and 5-6 PM, specifically on Monday and Wednesday morning to reduce delays. *[Data driven - §3.3.2 hour x weekday heat-map, PS-2]*
- Run a short review of what operational changes are done in between May and June, such as changes in staffing, machine downtime, supplier demand; to better understand the problem before introducing changes. *[Data driven - §3.3.3 monthly complaint trend, PS-2]*

### **4.3 Problem Statement 3 — Customer Retention and Behavioural Patterns**

**Interpretation :**

- The revenue is highly concentrated. Top 21.7% of customers generates 54.5% of total revenue by order frequency (4 + orders) and top 20% of customers generates 53.7% of revenue, (§3.4.1) shows a physically higher concentration than midterm's single vs regular split suggestion.
- Newly acquired customer repeat rate has fallen to 60.9% from 90.8% (March cohort to April cohort). This shows a retention problem is more in recent acquired customers, rather than just being a static 53.6% one-time visitor issue. (§3.4.2)
- Customers who raised a complaint on their first order is having a resolved rate of 89.9% which is significantly higher than customers with no complaints (61.2%), (§3.4.3) which shows complaint resolution is working properly and building a trust on customers.

**Recommendations :**

- Build a gentle push to customers who have ordered 2 or 3 times to buy again. There are 259 customers who makes up to 24.5% of revenue. Getting them to make 4-6 orders will largest revenue payoff per customer identified in the data. *[Data driven - §3.4.2 customer-tier Pareto, PS-3]*

- Use the WhatsApp or SMS mechanism for giving a reminder to the customers to return within 15-21 days. This approach is aiming the customers who have not returned within three weeks span. *[Data driven - §3.4.2 cohort analysis, PS-3]*
- Formalise the current complaint resolution process as a service guarantee and consider starting it explicitly to customer intakes, since (§3.4.3) shows that resolved complaints links to higher customer returns. *[Data driven - §3.4.3 complaint-resolution vs repeat analysis PS-3]*

## 4.4 Strategic Recommendations

The recommendations are not built from the data analysis. They (Structured Customer Feedback Form and Regular Customer Discount/Loyalty Tier) are came from general customer experience and market trends. These are strategic recommendations which are distinct from the above data driven suggestions.

### 4.4.1 Structured Customer Feedback Form

The business is currently just marking a complaint when it's occurred, and label them whether it's resolved/pending. There is no mechanism that tells what was the complaint. Also, there is no non-complaint feedback from the customers. A short structured feedback form is suggested to be sent to each customers by WhatsApp or SMS after every service. Also place the QR code to the form in-store/bills.

| Field                                    | Type                                | Purpose                                                           |
|------------------------------------------|-------------------------------------|-------------------------------------------------------------------|
| Customer ID / Name (optional)            | Short text                          | Link feedback to a customer record if needed                      |
| Service used                             | Dropdown (from the 11-service list) | Route feedback to the right service line                          |
| Overall satisfaction (1–5)               | Rating scale                        | Quick quantitative quality signal                                 |
| Was the item ready on the promised date? | Yes / No                            | Direct TAT-reliability check, complements §3.3.1                  |
| What could we improve?                   | Long text (optional)                | Spot service flaws, fix quality issues, and keep customers happy. |
| Would you recommend RDS Associates?      | Yes / No / Maybe                    | Simple loyalty/referral indicator                                 |

*Table 7: Feedback form template.*

This template is intended to provide to RDS associates and can publish it as a Google Form. This addresses a structural gap in the current dataset and get feedbacks from the customers and will help to improve the business performance. Place the QR of this form in-shop as well as in bills and

send the link in SMS/WhatsApp and add 'Reason for Complaint' and 'Suggestions' fields in the dataset.

#### **4.4.2 Regular-Customer Discount / Loyalty Tier**

We are able to analyse that regular customers are generating 78.44% orders and 79 % of total revenue. The new cohort data (§3.4.2) shows that, retention is weakening for more recently acquired customers.

Consider introducing a small percentage of discount or a free shoe cleaning (1-pair) offer to those customers after their 3rd or 4th order. It's a common mechanism in laundry and other repeat-service retail businesses to reward and reinforce existing loyalty, rather than a pattern discovered in this dataset.

### **4.5 Limitations**

- This is a four month (10 March 2026 to 10 July 2026) dataset. This length of the dataset is only sufficient to provide insights on short run stability and service level performance rather than confirming seasonal effects. That requires the full year for analysis.
- The 16 weeks revenue time series will be the descriptive level only. The longer series would need at least 30 observation points for the analysis. The collected data set does not capture the reasons behind the complaints. This would further limit the projects scope to identify the reason behind each complaints and scope reviewing the existing policies.

## **5. Presentation and Legibility of the Report**

This project follows a consistent structure and followed the rubrics. All sections and sub-sections were numbered and heading, sub-headings were in a standardised structure. Font style, paragraphing styles all are consistent. All the tables and figures are named, numbered and labelled immediately after the next paragraph that so that visual and narrative content stay aligned. Reference sections were mentioned in the required fields.